SYMATICS

Technical Overview

State-Resident Compute Architecture for Long-Context Inference and Multi-Agent Systems

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Current status: Gen1 FPGA validation in progress — first structural computing results demonstrated. Pre-silicon.

1. The Thesis: The Next Bottleneck Is State

In autoregressive inference, every generated token must re-read all active weights and state. As models move to long context and multi-agent workloads, the dominant cost is no longer arithmetic but data movement: weights stream from memory on every decode step, and context (KV cache) grows from megabytes to tens of gigabytes per sequence.

Symatics is building a state-resident compute architecture: silicon where information stays local, structured, and continuously available. The Morphific Processing Unit (MPU) keeps weights resident in local memory, bounds intermediate state traversal through a deterministic structural fabric, and fixes scheduling at compile time rather than runtime — so intelligence compounds across cycles instead of resetting on every request.

2. The MPU at a Glance

Key target specifications for the Gen2 device. These are design targets — not silicon-measured values.

Parameter	Target
Compute array	32,768 CUs (256 × 128), 16-bit FP16 datapath @ 1 GHz
Process node	TSMC N7, ≈ 430 mm², ≈ 32.8 B transistors
On-chip SRAM	384 MB (12 KB I-SRAM per CU)
External memory	4× GDDR7, ≈ 192 GB/s per chip
Interconnect	SIF fabric: 256 SerDes lanes/chip @ 16 Gbps; 64-port hub per domain
Package power	≈ 250–350 W, standard cold plate
Domain scale	64 chips per domain (~25 kW rack); compiler sees one (N×64×256) × 128 array

Scheduling is compile-time; there is no cache coherence, no runtime memory management, and no transaction layer in the interconnect — the PHY rate (16 Gbps) matches the compute rate (16-bit @ 1 GHz) by construction.

3. Validation Status: Gen1 on FPGA

Gen1 exists to prove the core principles of state-resident structural computing on real hardware before any tape-out commitment. Status as of October 2026:

Current progress

- FPGA prototype of the structural fabric and state paths is running
- Initial measurements of latency and state residency behavior completed
- Compiler-defined path scheduling demonstrated on the FPGA platform

Still in progress

> Full domain scaling (target 64-chip equivalent)

> Broader workload characterization

Reading guide: measured results on this page come from the Gen1 FPGA platform. All performance figures in Section 4 are analysis-derived projections, clearly labeled as such.

4. Performance, Stated Honestly

Single-stream decode throughput is bounded by the bandwidth wall: tokens/s ≈ memory bandwidth ÷ bytes per token. On dense 405B FP16 models, MPU and GPU hit the same wall — GPU HBM currently wins on bandwidth per dollar. The MPU case is conditional and specific:

Workload	Assessment	Basis
Cloud dense 405B serving	Not the wedge	Bandwidth wall; HBM wins per-dollar
MoE / sparse (DeepSeek-V3 class)	Core opportunity	11× fewer bytes/token than dense; structural routing
Long context (128K+)	Candidate opportunity	KV resident in local DRAM; no slow-storage descent while it fits
Edge / cost-sensitive inference	Secondary opportunity	Potential TOPS/W advantage if silicon delivers

At equal cost, analysis-derived competitiveness versus GPU is 0.27–0.54× on dense 405B FP16 and 0.54–0.90× on MoE (upper bound nets out the GPU all-to-all collective tax). These ranges are projections pending Gen1/Gen2 measurement — we publish them as analysis, not as benchmark results.

5. Roadmap

Generation	Goal	Status
Gen1 — architecture validation	Prove state-resident structural computing on FPGA; close the architecture	In progress — fabric running, first results demonstrated
Gen2 — production silicon	Tape-out with three-IC implementation (MPU + SIF hub + CSL); production MoE, long-context, agentic workloads	Planning; target of record set after Gen1 closure

Gen1 closes the architecture. Gen2 moves to production silicon.

6. Company & Contact

Symatics Technology Pte. Ltd. is a Singapore-headquartered fabless semiconductor startup with engineering in Singapore and Taiwan, founded and led by semiconductor veterans with 25+ years of experience across ATI/AMD, Lightelligence (HK:01879), Atmel, Western Digital, and HP. The team has shipped GPU, SoC, CPU, APU, PCIe IP, and flash-memory products from architecture through IPO.

Detailed specifications, measurement data, and architecture documents are available under NDA. Contact: **hello@symatics.sg** · Partnerships: **partnerships@symatics.sg** · Investors: **investors@symatics.sg** · **www.symatics.sg**

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